# Exponential Functions and Their Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

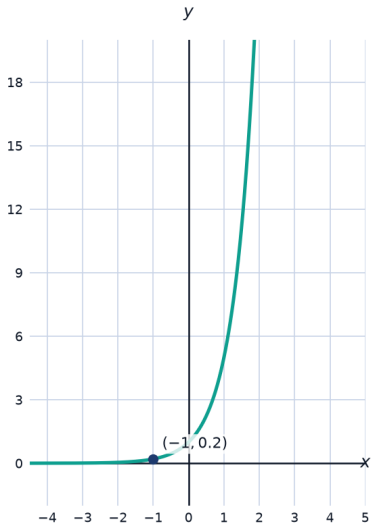
**1** [Foundational] • Target: Evaluating and describing

Let $f(x) = 3 \cdot 2^x$. Evaluate:

**a** $f(0)$ **b** $f(3)$ **c** $f(-2)$

**2** [Foundational] • Target: Evaluating and describing

For $y = 5^x$, state:




**a** the domain **b** the range

**c** the horizontal asymptote **d** the $y$-intercept

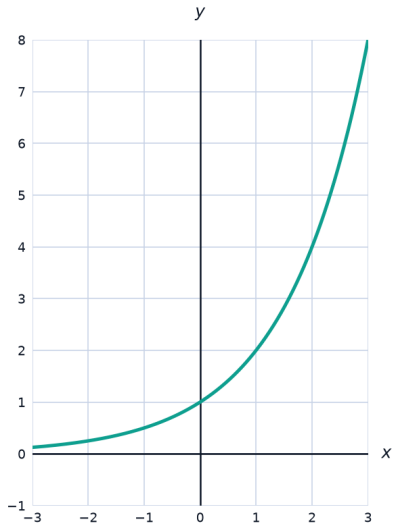
**3** [Foundational] • Target: Evaluating and describing

Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of $g$.

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**4** [Foundational] • Target: Identifying from graphs and applications

The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.

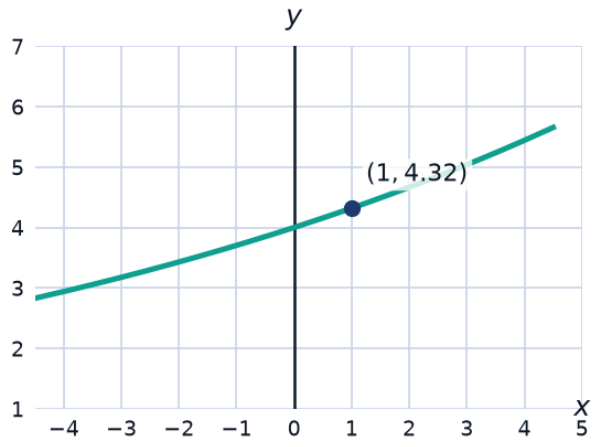



**5** [Foundational] • Target: Identifying from graphs and applications

An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%: $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.

**6** [Foundational] • Target: Identifying from graphs and applications

Determine whether each function represents growth or decay:




**a** $y = 4(1.08)^x$

**b** $y = 7(0.85)^x$

**c** $y = 2e^{-0.3x}$

**d** $y = \left(\frac{5}{4}\right)^x$

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**7** [Foundational] • Target: Compound interest

2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.

**a** Find the value after 3 years, to the nearest riyal.

**b** Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Compound interest

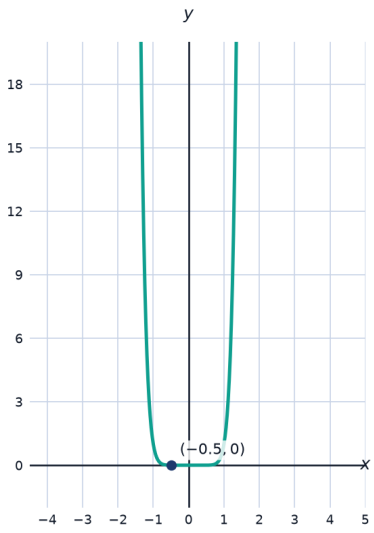
How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

**9** [Intermediate] • Target: Comparing exponential growth rates

Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.

**10** [Intermediate] • Target: Comparing exponential growth rates

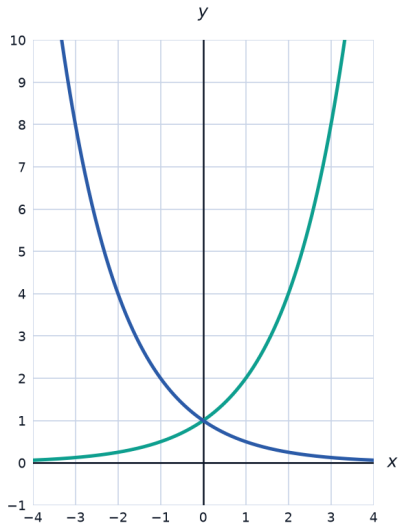
Rank the following functions from slowest to fastest growth as $x \to \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.




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**11** [Intermediate] • Target: Comparing growth and decay graphically

The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.




**12** [Intermediate] • Target: The natural exponential base e

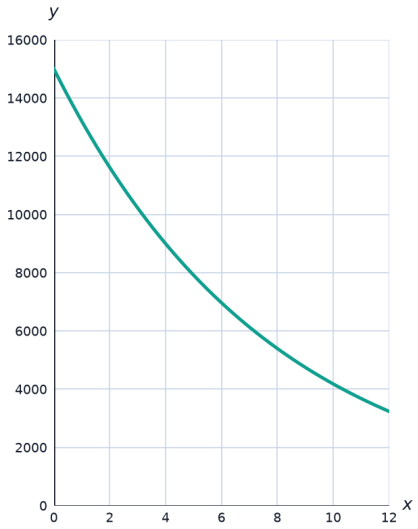
Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.

**13** [Intermediate] • Target: The natural exponential base e

Evaluate $e^0$, $e^1$ (to 3 decimal places), and simplify $e^{\ln 5}$.

**14** [Intermediate] • Target: Piecewise-exponential applications

A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).




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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

State the horizontal asymptote and range of each transformed exponential function:

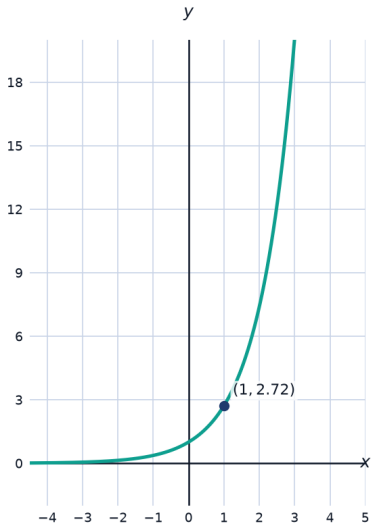
**a** $y = 3^x + 4$

**b** $y = -2^x$

**c** $y = 5^x - 3$

**16** [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Write the equation of the function obtained by taking $y = e^x$, reflecting it across the $y$-axis, then shifting it down 2 units.




**17** [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x + 1) = 3f(x)$.

**18** [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x - 1}{2x + 5}$.

**19** [Challenging] • Target: Evaluating and describing • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve the exponential equation $3^{2x - 1} = 5^{x + 2}$ for $x$ in terms of natural logarithms.

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**20** [Challenging] • Target: Identifying from graphs and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t) = 250e^{-0.048t}$, where $t$ is measured in days. Determine the exact daily multiplicative decay factor $b$ in the form $M(t) = 250b^t$, and calculate the daily percentage loss rounded to two decimal places.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Evaluating and describing

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

**a.** $A = Pe^{rt}$

**b.** $A = P/e^{rt}$

**c.** $A = P(1 + r)^t$

**d.** $A = Prt$

**22** [Foundational] • Target: Identifying from graphs and applications

The graph of $y = -2f(x + 1) - 3$ applies which transformations to $y = f(x)$?

**a.** Left 1, stretch 2, down 3 (no reflect)

**b.** Right 1, reflect, stretch 2, up 3

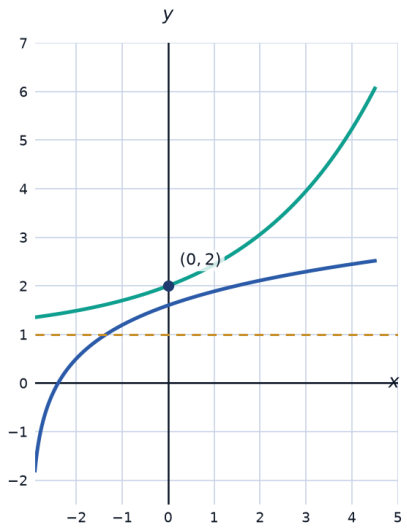
**c.** Right 1, stretch 2, down 3

**d.** Left 1, reflect over x-axis, vertical stretch 2, down 3

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**23** [Foundational] • Target: Compound interest

Express $\log_3 7$ using the change of base formula with natural log.




**a.** $\frac{\ln 7}{\ln 3}$

**b.** $7\ln 3$

**c.** $\frac{\ln 3}{\ln 7}$

**d.** $\ln 7 - \ln 3$

**24** [Foundational] • Target: Comparing exponential growth rates

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

**a.** $1$

**b.** $-1$

**c.** $2$

**d.** $-2$

**25** [Foundational] • Target: Comparing growth and decay graphically

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

**a.** 17

**b.** 13

**c.** 20

**d.** 16

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**26** [Foundational] • Target: The natural exponential base e

A scientist is modeling the population growth of a rare species of bioluminescent fungi. The population, $P(t)$, in thousands, after $t$ months is given by the function $P(t) = P_0 e^{kt}$, where $P_0$ is the initial population and $k$ is the growth constant. Observations show that after 2 months, the population has doubled from its initial size. Furthermore, after 6 months, the population is 16,000. What is the initial population $P_0$?

**a.** 2,000

**b.** 4,000

**c.** 8,000

**d.** 1,000

**27** [Foundational] • Target: Piecewise-exponential applications

Determine all continuous even functions $f$ satisfying $f(f(x)) = x$.

**a.** $f(x) = x^2$

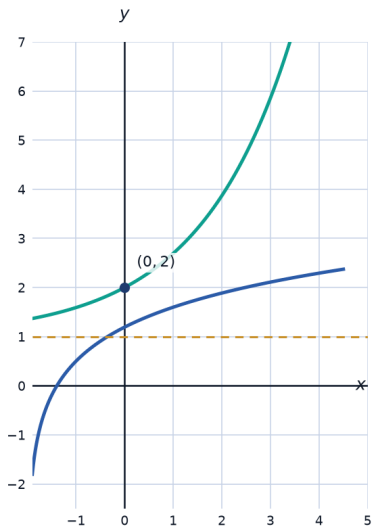
**b.** $f(x) = 1/x$

**c.** $f(x) = |x|$

**d.** $f(x) = x$ or $f(x) = -x$

**28** [Foundational] • Target: Transformations of exponential functions

Find the number of real-valued functions $f : \{1, 2, 3\} \to \{1, 2, 3\}$ satisfying $f(f(x)) = f(x)$ for all $x$.




**a.** 9

**b.** 10

**c.** 27

**d.** 6

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Comparing exponential functions numerically

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

**a.** $f^{-1}(x) = \frac{x-5}{2}$

**b.** $f^{-1}(x) = \frac{x+5}{2}$

**c.** $f^{-1}(x) = \frac{2}{x} - 5$

**d.** $f^{-1}(x) = 2x - 5$

**30** [Intermediate] • Target: Evaluating and describing

For $f : \mathbb{Z} \to \mathbb{Z}$, if $f(m+n) + f(mn-1) = f(m)f(n) + 1$ for all integers, find $f(2)$.

**a.** 1

**b.** 3

**c.** 4

**d.** 2

**31** [Intermediate] • Target: Identifying from graphs and applications

Consider the function $g(x) = x^2 - 6x + 5$. What is the x-coordinate of the vertex of the parabola represented by this function?

**a.** 3

**b.** -3

**c.** 6

**d.** -6

**32** [Intermediate] • Target: Compound interest

Solve: $f(x) = f(-x)$ for all $x$ where $f(x) = x^3 + ax$. Find $a$.

**a.** $a = -1$

**b.** Any $a$

**c.** $a = 0$

**d.** $a = 1$

**33** [Intermediate] • Target: Comparing exponential growth rates

Classify all solutions of $f(f(x)) = f(x)$ (idempotents) for $f(x) = x^2$.

**a.** $x = 1$ only

**b.** $x = 0$ or $x = 1$

**c.** All $x \geq 0$

**d.** $x = 0$ only

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**34** [Intermediate] • Target: Comparing growth and decay graphically

Given $f(x) = \frac{1}{2-x}$, find $f^{(n)}(x)$ (the $n$-th iterate).

**a.** $f^n(x) = nx + 1$

**b.** $f^n(x) = f(x)$ for all $n$

**c.** $f^n(x) \to \infty$

**d.** Cycles with period 3; $f^3(x) = x$

**35** [Intermediate] • Target: The natural exponential base e

Find all $k$ such that $f(x) = kx^2 + (k+2)x + 2k - 1$ is always negative.

**a.** $k < -1$

**b.** All $k < 0$

**c.** $k < -\frac{1}{3}$ and discriminant $< 0$

**d.** No such $k$ exists for all $x$

**36** [Intermediate] • Target: Piecewise-exponential applications

What is the half-life formula for exponential decay?

**a.** $A(t) = A_0(2)^{t/h}$

**b.** $A(t) = A_0 e^{-t}$

**c.** $A(t) = A_0/2^t$

**d.** $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$

**37** [Intermediate] • Target: Transformations of exponential functions

Find all real solutions to $f(2^x) = f(3^x)$ where $f(t) = t^2 - t$.

**a.** $x = 0$

**b.** $x = 1$

**c.** No solution

**d.** $x = \log_6 6$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Which is an exponential decay?

**a.** $y = 3^x$

**b.** $y = 0.4^x$

**c.** $y = e^x$

**d.** $y = 10^x$

**39** [Challenging] • Target: Evaluating and describing • Cross-Lesson: 3.03 Sum and Difference Formulas

Write using natural log: $\log_b x =$

**a.** $\frac{\ln b}{\ln x}$

**b.** $\ln x - \ln b$

**c.** $\ln x \cdot \ln b$

**d.** $\frac{\ln x}{\ln b}$

**40** [Challenging] • Target: Identifying from graphs and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

If $f(x) = \arcsin(\sin x)$, what is the period of $f$?

**a.** $2\pi$

**b.** $\pi$

**c.** $\pi/2$

**d.** Not periodic

**41** [Challenging] • Target: Compound interest • Cross-Lesson: 4.01 Circles and Parabolas

A function satisfies $f(1) = 2, f(2) = 5, f(3) = 10$. Identify the pattern.

**a.** $f(n) = n + 1$

**b.** $f(n) = 2^n$

**c.** $f(n) = n^2 + 1$

**d.** $f(n) = 3n - 1$

**42** [Challenging] • Target: Comparing exponential growth rates • Cross-Lesson: 4.02 Ellipses

Find the range of $f(x) = \frac{1}{x^2 + 1}$ for all real $x$.

**a.** $(0, 1)$

**b.** $(0, 1]$

**c.** $[0, 1]$

**d.** $[0, 1)$

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**43** [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 4.03 Hyperbolas

Given $f(x) = x^n$ for integer $n \geq 1$, find the number of real solutions to $f(f(x)) = x$.

**a.** Always $n$

**b.** Depends on $n$: $n^2$ solutions in general

**c.** Always 1

**d.** Always 2

**44** [Challenging] • Target: The natural exponential base e • Cross-Lesson: 5.01 Vectors: Components and Operations

Find all $x$ satisfying $f(x) > 0$ where $f(x) = (x-1)(x-2)(x-3)$.

**a.** 13

**b.** $x < 1$ or $2 < x < 3$

**c.** $x > 1$

**d.** $x < 3$ only

**45** [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 5.02 The Dot Product

If $f$ is monotonically increasing, prove $f^{-1}$ is also increasing.

**a.** True only if $f$ is linear

**b.** True only if $f$ is differentiable

**c.** If $a < b$ then $f^{-1}(a) < f^{-1}(b)$ — otherwise $f(f^{-1}(a)) \geq f(f^{-1}(b))$ contradicting monotonicity

**d.** Not necessarily true

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